

HAT-P-12b

R-0666

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_260423 · created 2026-07-27T18:04:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461153.7744 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.021
Transit depth (Rp/Rs)^2	2.2 +/- 0.63 [%]
Orbital Inclination (inc)	86.4 +/- 1.25
Airmass coefficient 1 (a1)	1.198 +/- 0.019
Airmass coefficient 2 (a2)	-0.18 +/- 0.12
Scatter in the residuals of the lightcurve fit is	1.58 %
Best Comparison Star	#3 - [121, 379]
Optimal Aperture	3.37
Optimal Annulus	7.34
Transit Duration (day)	0.07 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.148 $\pm$ 0.021 vs 0.1406 (deviation 0.32 σ)
Duration fit vs published	0.07 d vs 0.0975 d (deviation 1.05 σ)
Mid-time O-C	12.9 min
Depth SNR	6.4
Residual scatter	1.58 %

Light curve

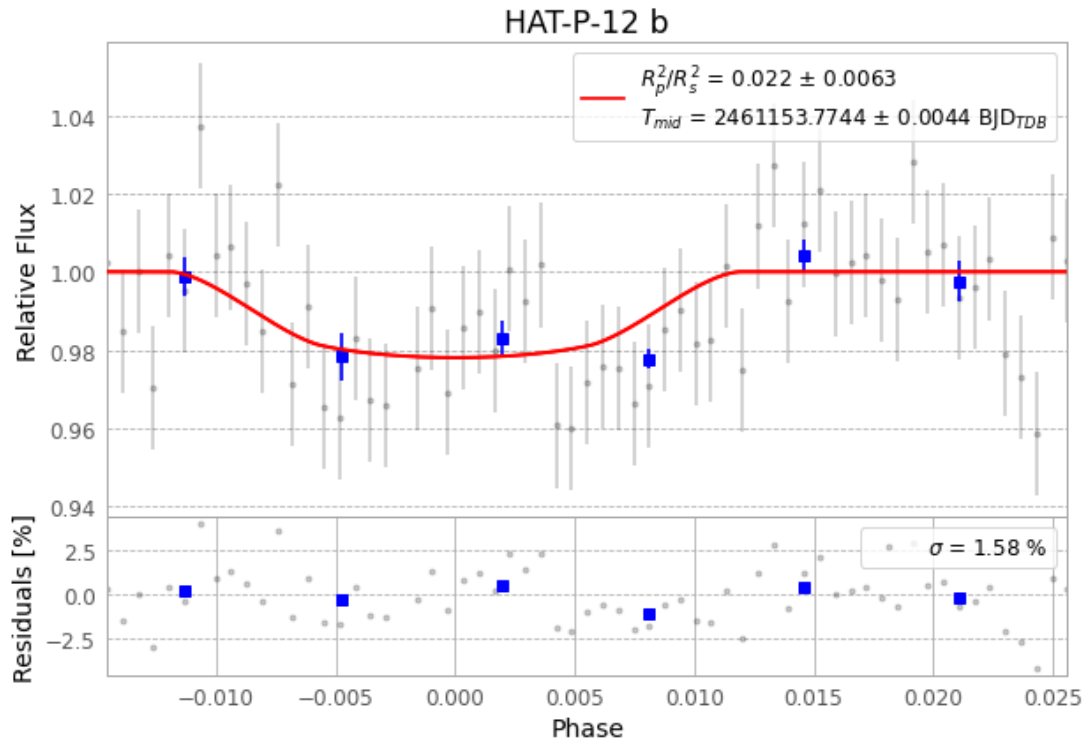


Figure 1 — FinalLightCurve_HAT-P-12 b_2026-04-22.png (SHA-256 8442ac625b07d89f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [282, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ebc89b7087ed095f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d476b1b

Data provenance

63 FITS frames (41 MB), HATP-12260423052115.FITS → HATP-12260423082716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-260423.log (6aec407665cc...), FinalLightCurve_HAT-P-12 b_2026-04-22.png (8442ac625b07...), FinalLightCurve_HAT-P-12 b_2026-04-22.csv (63646402cd87...)

Compute resources

Wall time 105.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 18:04Z.